UNIT - II
Flanged Beams, Bond, Shear & Torsion

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Flanged Beams (T and L Beams)

Why flanged beams?

- In a monolithic RC floor, the **slab and beam cast together** act as one unit. A part of the slab on each side of the web works as the **flange** and takes compression.
- Cross-section is **T-shaped** (interior beam, slab both sides) or **L-shaped** (edge beam, slab on one side).
- Advantages: large compression area, so **less steel**, smaller depth, economical for long spans.

In a flanged beam the flange is in compression only in **sagging (span) zones**. Near supports (hogging) the flange is in tension, so the beam is designed as a **rectangular beam of width b_w**.

Slab (flange) b_f x D_f → Web b_w x (D - D_f) → Tension steel A_{st} at bottom

Effective flange width (IS 456 cl. 23.1.2)

Beam	Effective width b_f
T-beam (continuous or simply supported)	$l_o/6 + b_w + 6\,D_f$
L-beam (continuous or simply supported)	$l_o/12 + b_w + 3\,D_f$
Isolated T-beam	$l_o / (l_o/b + 4) + b_w$
Isolated L-beam	$0.5\,l_o / (l_o/b + 4) + b_w$

- l_o = distance between points of zero moment (simply supported span = effective span; continuous = about 0.7 x span).

- b_w = web width, D_f = flange (slab) thickness, b = actual width of flange.
- **Rule:** b_f must never exceed the actual centre-to-centre spacing of the beams.

Take the **least** of the calculated b_f and the actual spacing. Memorise: T $\longrightarrow l_o/6 + b_w + 6D_f$; L $\longrightarrow l_o/12 + b_w + 3D_f$.

Position of neutral axis

- Assume the NA at the **bottom of flange**, $x_u = D_f$, and find the moment of resistance of the flange:

$$M_{u,f} = 0.36\, f_{ck}\, b_f\, D_f\, (d - 0.42\, D_f)$$

- If $M_u <= M_{u,f}$: NA lies in the flange ($x_u <= D_f$). Design as a **rectangular beam of width b_f**.
- If $M_u > M_{u,f}$: NA lies in the web ($x_u > D_f$). Use the flanged-beam equations below.

Calculate b_f → Find $M_{u,f}$ ($x_u = D_f$) → Compare M_u with $M_{u,f}$ → Choose case 1 or case 2

Case 1: Neutral axis within flange ($x_u <= D_f$)

- Section behaves as a **rectangular section b_f x d**.

$$M_u = 0.87\, f_y\, A_{st}\, d\, (1 - A_{st}\, f_y / (b_f\, d\, f_{ck}))$$

- Depth of NA: $x_u = 0.87\, f_y\, A_{st} / (0.36\, f_{ck}\, b_f)$; check $x_u <= D_f$ and $x_u <= x_{u,max}$.
- Limiting values $x_{u,max}/d$: **0.53 (Fe250), 0.48 (Fe415), 0.46 (Fe500)**.

Case 2: Neutral axis in web ($x_u > D_f$)

Compression is taken by the web (rectangular stress block above NA) plus the **projecting flange**.

- **(a) When $D_f/d <= 0.2$** (thin flange, stress is nearly uniform at 0.447

f_{ck}):

$$M_u = 0.36\, f_{ck}\, b_w\, x_u\, (d - 0.42\, x_u) + 0.45\, f_{ck}\, (b_f - b_w)\, D_f\, (d - D_f/2)$$

$$0.87\, f_y\, A_{st} = 0.36\, f_{ck}\, b_w\, x_u + 0.45\, f_{ck}\, (b_f - b_w)\, D_f$$

- **(b) When $D_f/d > 0.2$** (thick flange, stress is not uniform), replace D_f by the equivalent depth y_f:

$$y_f = 0.15\, x_u + 0.65\, D_f \text{ (but } y_f <= D_f)$$

$$M_u = 0.36\, f_{ck}\, b_w\, x_u\, (d - 0.42\, x_u) + 0.45\, f_{ck}\, (b_f - b_w)\, y_f\, (d - y_f/2)$$

$$0.87\, f_y\, A_{st} = 0.36\, f_{ck}\, b_w\, x_u + 0.45\, f_{ck}\, (b_f - b_w)\, y_f$$

Procedure for $x_u > D_f$: assume x_u ⟶ find A_{st} from force equilibrium ⟶ compute M_u ⟶ adjust. Always check $x_u <= x_{u,max}$ (else beam is over-reinforced).

Design steps for a T-beam

Fix span, loads, fck, fy → Assume D = 1/12 to 1/15 and d, b_w → Find M_u = 1.5 M and b_f → Check $M_{u,f}$ (NA position) → Find A_{st}, verify x_u → Detail steel, check shear, deflection

- Ast limits: $A_{st,min} = 0.85\, b_w\, d / f_y$, $A_{st,max} = 0.04\, b_w\, D$.

Worked Example 1: Design of a T-beam

Given: simply supported T-beam, effective span 6 m, spacing of beams 3 m c/c, b_w = 300 mm, D_f = 100 mm, d = 500 mm, M25 and Fe415, factored moment M_u = 500 kN m. Find A_{st}.

- **Step 1 - flange width:** $b_f = l_o/6 + b_w + 6\,D_f = 6000/6 + 300 + 600 =$ **1900 mm** < 3000 mm spacing, so OK.
- **Step 2 - NA position:** $M_{u,f} = 0.36 \times 25 \times 1900 \times 100 \times (500 - 42) =$ **783 kN m**.
- Since $M_u = 500 < 783$: **NA lies in the flange**, so design as rectangular $b_f = 1900$, $d = 500$.
- **Step 3 - steel:** $500 \times 10^6 = 0.87 \times 415 \times A_{st} \times 500\,(1 - 415\,A_{st} / (1900 \times 500 \times 25))$.
- Simplify: $A_{st}^2 - 57229\,A_{st} + 1.585 \times 10^8 = 0$, so **$A_{st}$ = 2918 mm²**.
- **Step 4 - check:** $x_u = 0.87 \times 415 \times 2918 / (0.36 \times 25 \times 1900) =$ **61.6 mm < D_f** (OK) and < $x_{u,max} = 0.48 \times 500 = 240$ mm.
- $A_{st,min} = 0.85 \times 300 \times 500 / 415 = 307$ mm² < provided.

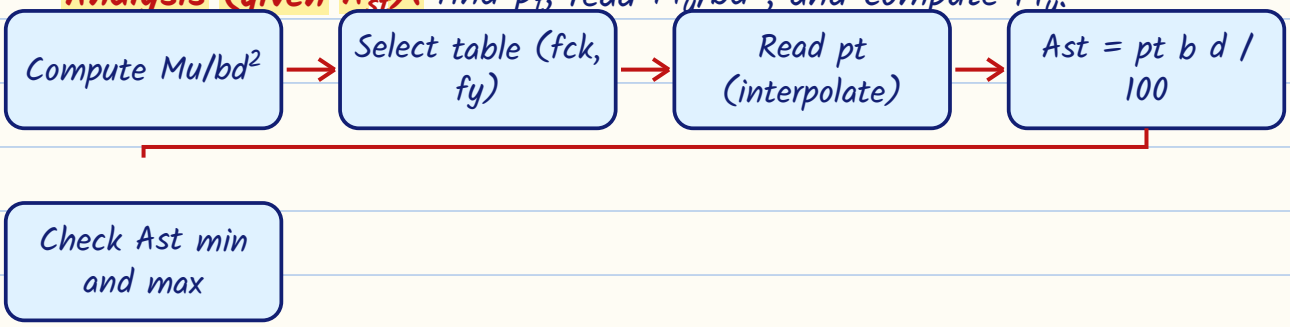
Provide 6 bars of 25 mm dia (A_{st} = 2945 mm²) in two layers, (3 + 3). Nominal cover 25 mm for mild exposure; add stirrups and check shear and deflection.

2. Use of Design Aids for Flexure (SP-16)

- **SP-16** (Design Aids for IS 456) gives tables and charts that avoid trial and error in flexure design.
- The tables give **$M_u/(b\,d^2)$** against **$p_t = 100\,A_{st}/(b\,d)$** for each grade of concrete.

SP-16 item	Content
Tables 1 to 4	Rectangular sections, singly reinforced: M_u/bd^2 vs p_t for fy 250, 415, 500
Tables 45 to 56	Doubly reinforced sections, with d'/d ratios
Tables 57 to 59	**Flanged beams**: $M_u/(b_w\,d^2)$ for varying b_f/b_w and D_f/d
Charts 1 to 18	Same data presented as curves

- **Procedure for design (given M_u, b, d):**
  - Compute $M_u / (b\ d^2)$ (in N/mm^2).
  - From the table for the given f_{ck} and f_y, read p_t (interpolate between rows).
  - $A_{st} = p_t\ b\ d / 100$.
- **Analysis (given A_{st}):** find p_t, read M_u/bd^2, and compute M_u.



Design aids give quick, reliable answers (accurate to about 1-2 %). For flanged beams find b_f, then check D_f/d and b_f/b_w to enter Tables 57-59.

3. Behaviour of RC Members in Bond and Anchorage

What is bond?

- **Bond** is the adhesion and grip between concrete and reinforcing steel which makes them act together (strain compatibility).
- Bond comes from three sources: **chemical adhesion**, **friction**, and **mechanical interlocking** of ribs (the most important for deformed bars).
- **Bond stress** tau_b = shear stress at the steel-concrete interface = change in bar force / (perimeter x length).
- Bond failure types: pull-out of the bar, splitting of the concrete cover (more common).

Design bond stress tau_{bd} (IS 456 cl. 26.2.1.1)

Grade of concrete	M20	M25	M30	M35	M40 and above
tau_{bd} for plain bars (N/mm^2)	1.2	1.4	1.5	1.7	1.9

Deformed bars (60 % more)	1.92	2.24	2.4	2.72	3.04

- For bars in **compression**, τ_{bd} is **increased by 25 %**.
- Deformed bars (HYSD, Fe415/500) give **60 % higher** τ_{bd} than plain bars.

Development length L_{d}

Length of bar required to develop its design stress ($0.87 f_y$) by bond.

$$L_d = \phi \sigma_s / (4 \tau_{bd}) = 0.87 f_y \phi / (4 \tau_{bd})$$

- ϕ = nominal diameter of bar, $\sigma_s = 0.87 f_y$ = stress in bar at section.
- **Fe415, M20, deformed:** $L_d = 0.87 \times 415 \phi / (4 \times 1.92) = 47 \phi$.
- Typical values for Fe415 deformed tension bars: M20 ⟶ 47 phi, M25 ⟶ 40 phi, M30 ⟶ 38 phi.

Use L_d for **anchoring** a bar beyond the section of maximum stress.
Hooks are not allowed for compression bars.

Anchorage by bends and hooks

Type	Anchorage value	Notes
Standard 180 deg hook (U-type)	16 phi	straight extension >= 4 phi (min 65 mm)
Standard 90 deg bend	8 phi	straight extension >= 4 phi
Min internal radius of bend	2 phi (mild steel), 4 phi (HYSD)	to avoid concrete crushing

Check for development length at supports (cl. 26.2.3.3)

For a **positive-moment (sagging) bar** at simple supports:

$$L_d <= 1.3 M_1 / V + L_o$$

- M_1 = moment of resistance of the section assuming all bars at the

section are stressed to 0.87 f_y.

- **V** = shear force at the section due to design loads; L_o = anchorage beyond the centre of support (equivalent anchorage of hook/bend). L_o is limited to **d or 12 phi, whichever is greater**.
- Factor **1.3** is used when the end of the reinforcement is confined by a compressive reaction (support).
- At the point of inflexion, **L_d <= M_1/V + L_o**.

This check controls the size of bars at simple supports: smaller diameters have smaller L_d, so choose closely spaced small bars.

Laps, splices and curtailment

- **Lap length in flexural tension >= L_d or 30 phi** (whichever is greater); compression >= L_d or 24 phi.
- Direct tension: lap >= 2 L_d or 30 phi. Do not lap bars larger than **36 mm** dia (use mechanical/welded splices).
- Laps should be staggered, away from points of maximum stress. Not more than **50 %** bars lapped at a section.
- **Curtailment:** each bar extends beyond its theoretical cut-off point by **d or 12 phi** (greater). At least **1/3** of positive steel reaches the support (simple support) and 1/4 for continuous.
- Never curtail in a tension zone unless shear is small (V_u <= 2/3 of shear resistance) or extra stirrups are provided.

Worked Example 2: Check for development length

Given: beam 300 x 550 (d = 500), 3 bars of 20 mm dia (A_{st} = 942 mm^2) continue to a simple support, M20 and Fe415, factored shear at support V = 180 kN. Check the development length.

- tau_{bd} = 1.2 x 1.6 = 1.92 N/mm^2 (deformed bars). L_d = 0.87 x 415 x 20 / (4 x 1.92) = **940 mm**.
- M_1 = 0.87 f_y A_{st} d (1 - A_{st} f_y/(b d f_{ck})) = 361 x 942 x 500 (1 - 0.13) = **148 kN m**.

- $M_1/V = 148 \times 10^6 / 180000 =$ **822 mm**. With support width 230 mm: $L_o = 115 - 25 = 90$ mm.
- Limit = $1.3 \times 822 + 90 =$ **1158 mm** > $L_d = 940$ mm. Hence **OK, safe in bond.**

If the check fails: use smaller diameter bars (more of them), or provide a standard 90 deg bend / hook, or increase support width.

4. Behaviour of RC Beams in Shear

Diagonal tension

- Shear force V in a beam produces shear stress; combined with flexural stress it creates **principal tensile stress** inclined at about 45 deg.
- Concrete is weak in tension, so **diagonal tension cracks** form near supports, starting at the NA and spreading to the tension face.
- Shear failure is **sudden and brittle**, hence shear steel must be provided so that the beam fails first in flexure.

Modes of shear failure

Mode	Reason	Remarks
Diagonal tension failure	a/d = 2.5 to 6, little web steel	Sudden, inclined crack to compression zone
Shear compression failure	a/d = 1 to 2.5	Crushing of concrete above the crack head
Shear tension (web crushing)	thin-webbed I beams	Crushing of web by diagonal compression

- a = shear span (distance of load from the support). Deep beams (a/d < 1) behave as **tied arch**.

Nominal shear stress

$$\tau_v = V_u / (b\ d)$$

- $b = b_w$ for flanged beams. For beams of varying depth: $\tau_v = (V_u \pm M_u \tan(\beta)/d) / (b\,d)$.
- **Critical section** for shear: at a distance **d from face of support** (when support reaction produces compression in the end region).
- The design shear strength of concrete **τ_c** depends on the percentage of tension steel $p_t = 100\,A_{st}/(b\,d)$ and f_{ck}.

Table 19 - τ_c (N/mm^2), selected values

p_t (%)	0.25	0.50	0.75	1.00	1.50	2.00	3.00 and above
M20	0.36	0.48	0.56	0.62	0.72	0.79	0.82
M25	0.36	0.49	0.57	0.64	0.74	0.82	0.92
M30	0.37	0.50	0.59	0.66	0.76	0.84	0.96

- For $p_t < 0.15$ % take τ_c for 0.15 % (M20: 0.28). **Interpolate** for intermediate p_t.
- A_{st} = area of **tension steel continuing at least d beyond the section** and properly anchored.

Table 20 - maximum shear stress $\tau_{c,max}$ (N/mm^2)

Grade	M15	M20	M25	M30	M35	M40 and above
$\tau_{c,max}$	2.5	2.8	3.1	3.5	3.7	4.0

If **$\tau_v > \tau_{c,max}$**, the section must be **redesigned** (increase b or d or grade of concrete); shear steel cannot cure web crushing.

Design of shear reinforcement (cl. 40.4)

- **Case (i): $\tau_v \le \tau_c$** - no calculation needed, but provide **minimum shear reinforcement**.
- **Case (ii): $\tau_c < \tau_v \le \tau_{c,max}$** - design stirrups for the excess shear:

$$V_{us} = V_u - \tau_c\, b\, d$$

$$s_v = 0.87 f_y A_{sv} d / V_{us}$$

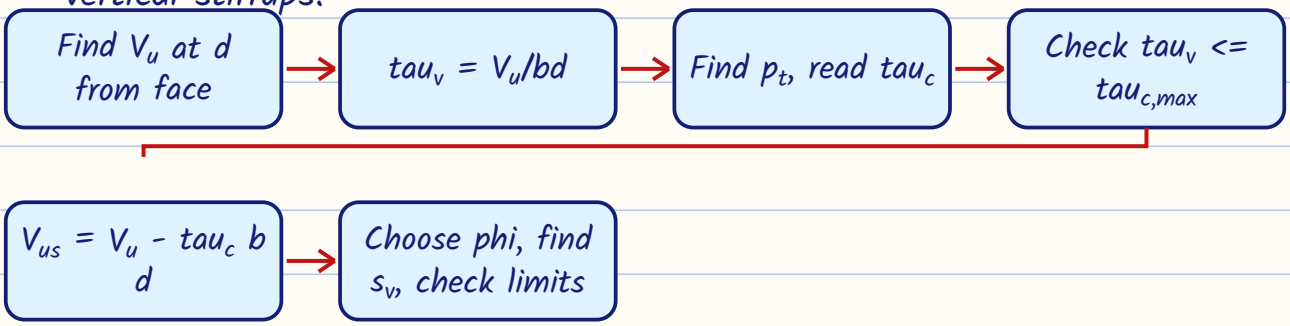
- A_{sv} = total area of stirrup legs (2-legged: 2 x pi/4 x phi^2); s_v = spacing.
- **Minimum shear reinforcement:**

$$A_{sv} / (b s_v) >= 0.4 / (0.87 f_y)$$

- **Max spacing of stirrups:** the least of **0.75 d** (vertical), **d** (inclined), **300 mm**.
- Where $\tau_v > \tau_c$, stirrup size is usually 6, 8 or 10 mm; spacing is increased towards midspan.

Bent-up bars

- Shear resistance of bent-up bars: $V_{us} = 0.87 f_y A_{sb} \sin(\alpha)$, where alpha = angle of bent (45 deg usually).
- Bent-up bars should **not carry more than 50 %** of V_{us}; the rest is by vertical stirrups.



Worked Example 3: Design of shear reinforcement

Given: simply supported beam b = 300, d = 500, M20, Fe415. Effective span 6 m, factored load w_u = 60 kN/m. Tension steel at support = 3 bars of 20 mm (942 mm^2).

- Support shear V_u = 60 x 6/2 = 180 kN. At d from face (about 0.5 m from centre): **V_u = 180 - 60 x 0.5 = 150 kN**.
- τ_v = 150 x 1000/(300 x 500) = **1.0 N/mm^2** < $\tau_{c,max}$ = 2.8, OK.
- p_t = 942 x 100/(300 x 500) = 0.63 %: τ_c = 0.48 + (0.13/0.25) x 0.08 = **0.52 N/mm^2**.

- Since $tau_v > tau_c$: $V_{us} = 150000 - 0.52 \times 300 \times 500 =$ **72 kN**.
- Use 8 mm dia 2-legged stirrups: $A_{sv} = 100.6\ mm^2$.
- $s_v = 0.87 \times 415 \times 100.6 \times 500 / 72000 =$ **252 mm**.
- Max spacing: $0.75\ d = 375$; 300; minimum steel limit $s_v <= 0.87\ f_y\ A_{sv} / (0.4\ b) = 303$ mm.

Provide 8 mm 2-legged stirrups @ 250 mm c/c near supports. In the middle region where V_u is small use the minimum shear reinforcement (@ 300 mm c/c).

5. Behaviour of RC Beams in Torsion

Basics

- **Torsion** is the twisting moment T acting about the longitudinal axis. It arises in **edge beams, spandrel beams, curved beams, canopy and balcony beams, and beams supporting eccentric loads**.
- Two types: **Equilibrium torsion** (statically determinate; must be designed for) and **compatibility torsion** (from continuity; can be redistributed, usually minimum nominal steel).
- Torsion produces **diagonal tension cracks in a spiral** around the member: it needs **closed stirrups + longitudinal bars at corners**.

Torque T → Shear flow around section → Diagonal tension spiral cracks → Closed stirrups + longitudinal bars

IS 456 method (cl. 41): equivalent shear and moment

Torsion is converted to an **equivalent shear V_e** and **equivalent bending moment M_e**, then designed as for flexure and shear.

$$V_e = V_u + 1.6\ T_u / b$$

$$M_t = T_u\ (1 + D/b) / 1.7$$

- **$M_{el} = M_u + M_t$** (for the tension face, bottom steel in sagging).
- **$M_{e2} = M_t - M_u$** (acts on the opposite face; provided only when $M_t > M_u$, steel on the compression side).
- T_u = factored torque, b = width, D = overall depth.

$$\tau_{ve} = V_e / (b\, d) \text{ must not exceed } \tau_{c,max}$$

Transverse and longitudinal reinforcement

- **Transverse (closed stirrups):**

$$A_{sv} / s_v = T_u / (b_1\, d_1\, 0.87\, f_y) + V_u / (2.5\, d_1\, 0.87\, f_y)$$

- b_1, d_1 = centre-to-centre distance between corner bars (width and depth). The total should not be less than **$(\tau_{ve} - \tau_c)\, b\, / (0.87\, f_y)$**.
- **Stirrup spacing** is the least of **x_1, $(x_1 + y_1)/4$, 300 mm** (x_1, y_1 = short and long dimensions of closed stirrup).
- **Longitudinal steel** is designed for M_{el} (bottom) and M_{e2} (top).
- **Side face steel:** for $D > 450$ mm provide **0.1 % of web area** ($b \times D$) distributed equally on two faces, spacing ≤ 300 mm.

> Torsion steel is **additional** to flexure steel but shear steel in torsion is designed using equivalent V_e; it is not added to the shear stirrups.

Worked Example 4: Beam under bending, shear and torsion

Given: beam 300 x 500 mm ($d = 450$, cover 25 mm), M25 and Fe415.

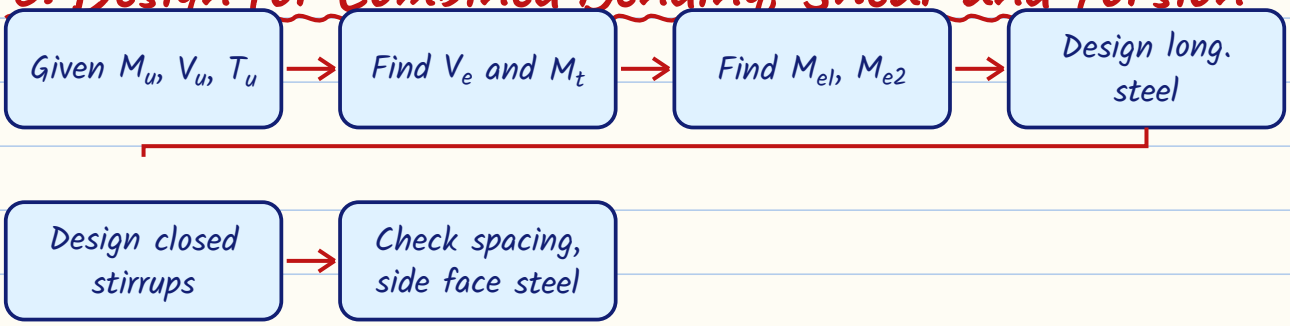
Factored loads: $M_u = 80$ kN m, $V_u = 60$ kN, $T_u = 20$ kN m.

- **V_e = 60 + 1.6 x 20/0.3 = 166.7 kN**; τ_{ve} = 166700/(300 x 450) = **1.235 N/mm²** < 3.1 (OK).
- **M_t** = 20 (1 + 500/300)/1.7 = **31.4 kN m**; M_{el} = 80 + 31.4 = **111.4 kN m**. M_{e2} = 31.4 - 80 < 0, so no top steel by torsion.
- $M_{u,lim} = 0.138 \times 25 \times 300 \times 450^2 = 210$ kN m > M_{el}: singly reinforced. $M_{el}/bd^2 = 1.83$, so **p_t = 0.56 %**, **A_{st} = 756 mm²** (provide 4-16 mm = 804 mm²).

- tau_c (p_t = 0.56 %, M25) = 0.51; minimum A_{sv}/s_v = (1.235 - 0.51) x 300/361 = **0.60 mm^2/mm**.
- b_1 = 300 - 2 x 25 - 2 x 10 - 16 = 214; d_1 = 414 mm. A_{sv}/s_v = 20 x 10^6/(214 x 414 x 361) + 60000/(2.5 x 414 x 361) = 0.625 + 0.161 = **0.79 mm^2/mm**.
- 10 mm 2-legged (157 mm^2): s_v = 199 mm; max spacing = min(214, (214 + 414)/4 = 157, 300) = 157.

Provide 10 mm 2-legged closed stirrups @ 150 mm c/c, 4-16 mm at bottom, 2-12 mm at top as hangers, and side face steel (D = 500 > 450): 0.1 % x 300 x 500 = 150 mm^2, i.e. 1-12 mm on each face.

6. Design for Combined Bending, Shear and Torsion



- A beam under **bending + shear + torsion** is designed by IS 456 cl. 41.4 using **equivalent loads**.
- **Step 1:** compute V_e and tau_{ve}; check tau_{ve} <= $tau_{c,max}$ (else redesign the section).
- **Step 2:** compute M_{el} (and M_{e2}); design the longitudinal steel.
- **Step 3:** transverse steel from torsion + shear formula; take the larger from the two checks.
- **Step 4:** detail: closed stirrups with 135 deg hooks, corner bars, side bars, correct anchorage.
- If **tau_{ve} < tau_c**, only **minimum shear reinforcement** is required, torsion steel is not needed.

7. Serviceability: Deflection and Cracking

Limit state of serviceability

- Beams must be **safe in strength** and also **serviceable**: no excessive deflection or cracking that hurts appearance or function.

Control of deflection (cl. 23.2)

- IS 456 allows the **span/effective depth ratio** check instead of calculating deflection.
- Basic values of l/d: **Cantilever 7, simply supported 20, continuous 26**. (For spans > 10 m: multiply by 10/span except cantilevers.)
- Final allowable l/d = **basic value x k_t x k_c x k_f**.

Factor	Meaning
k_t	Modification for **tension steel**; depends on percentage steel and stress $f_s = 0.58\, f_y\, (A_{st,req}/A_{st,prov})$
k_c	Modification for **compression steel** (p_c); up to 1.5
k_f	Flanged beams: 0.8 for $b_w/b_f <= 0.3$, interpolate to 1.0 at $b_w/b_f = 1$

- Total deflection limits: **span/250** (final, under all loads), and **span/350 or 20 mm** after finishing (that part affecting partitions and finishes).

Rule of thumb: if actual l/d <= allowable l/d, deflection is OK. Otherwise **increase depth d** rather than steel.

Control of cracking (cl. 26.3 and 35.3.2)

- Cracking is **inevitable** in tension zone of RC. It is controlled by limiting **crack width** and **bar spacing**.
- Limits of crack width: **0.3 mm** (normal), 0.2 mm (moderate/severe exposure) in IS 456 cl. 35.3.2.
- Use **closely spaced small diameter bars** rather than few large bars.
- Clear distance between parallel bars: not less than **bar dia, or max size of aggregate + 5 mm, or 20 mm**.
- Max horizontal spacing of tension bars: **not more than 300 mm** for

Fe250 and about **180 mm (Fe415) / 150 mm (Fe500)** per Table 15.

Quick Revision

- b_f: $T = l_o/6 + b_w + 6D_f$; $L = l_o/12 + b_w + 3D_f$.
- If $M_u <= 0.36\ f_{ck}\ b_f\ D_f\ (d - 0.42\ D_f)$, design as rectangular b_f x d.
- If $D_f/d > 0.2$, use $y_f = 0.15\ x_u + 0.65\ D_f$.
- $L_d = 0.87\ f_y\ phi\ 1/(4\ tau_{bd})$; check $L_d <= 1.3\ M_1/V + L_o$.
- Shear: $tau_v = V_u/bd$; $V_{us} = V_u - tau_c\ b\ d$; $s_v = 0.87\ f_y\ A_{sv}\ d/V_{us}$.
- Torsion: $V_e = V_u + 1.6\ T/b$; $M_t = T(1 + D/b)/1.7$; $M_{el} = M_u + M_t$.
- Deflection: l/d = 7, 20, 26 x modification factors.

Important Questions

- 1. Derive the expression for M_u of a T-beam when the NA is in the web and $D_f/d > 0.2$.
- 2. Explain effective flange width of T and L beams as per IS 456.
- 3. Define bond. Explain development length and the check at simple supports.
- 4. Explain the modes of shear failure of RC beams. How is shear reinforcement designed?
- 5. A T-beam has $b_f = 1200$, $b_w = 250$, $D_f = 100$, $d = 500$. Find A_{st} for $M_u = 350$ kN m.
- 6. Explain the IS 456 method of design for combined bending, shear and torsion.
- 7. Write notes on the control of deflection and cracking in beams.